Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– Ho Chi Minh, September 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
09/09/26	A	ThongLQ	Created Project Management Plan based on Report 1 and SEP490 execution schedule.
09/09/26	A	NghiaHQ	Added scope estimation, quality objectives and project risks.
09/09/26	A	KhanhGNG	Added process, deliverables and quality/testing activities.
09/09/26	A	DucDDA	Added training, communication, configuration management and infrastructure plan.
12/09/26	M	Project team	Revised working plan: four main flows, external AI API scope, flow ownership, testing and risks; aligned dates with Project Schedule and standardized tables.
12/09/26	M	Project team	Aligned with supplied Sep 2026 council roadmap: common dossier deadlines, W6 scope freeze, W7 Package 1, W8-W9 Review 2, W14 eligibility and W15 defense; recurring tracking, submission lead times and conditional Council 2. Dates remain provisional.
16/09/26	M	Project team	Aligned flow ownership and existing cross-cutting assignments with the team decision; updated existing Web/database/version-control entries to Next.js, Supabase PostgreSQL and GitHub. No additional setup or Auth feature added.
19/10/2026	M	Project team	Added Review 2 readiness gates, mission update decision, source/design gaps and unchanged-registration scope clarification.

*A - Added M - Modified D - Deleted

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

DroneOps Intelligence retains the ten Major Features in Report 1, organized into four main flows. MF1: Mission Planning & Pre-flight Validation (FE-02/03/04); MF2: Real-Time Monitoring & Incident Management (FE-05/06/07 and incident functions of FE-08/10); MF3: Post-Flight Analysis & Mission History (FE-08/10); MF4: AI-Assisted Mission Risk Assessment (FE-09/10). FE-01 supports all flows. Complexity reflects integration, business rules and real-time processing. Effort remains an initial planning estimate to be refined after the SRS/design baseline.

#	WBS Item	Complexity	Est. Effort (man-days)
1	FE-01 UAV & Operator Management	Medium	12
2	FE-02 Mission Planning & Management	Complex	20
3	FE-03 Mission Approval Workflow	Medium	12
4	FE-04 Geofence & Flight Path Validation	Complex	24
5	FE-05 MAVLink Telemetry Integration	Complex	24
6	FE-06 Real-Time Mission Monitoring	Complex	20
7	FE-07: Operational Alert and Incident Detection	Complex	18
8	FE-08 Incident Management & Post-Flight Analysis	Medium	14
9	FE-09 AI-Assisted Mission Decision Support	Medium	24
10	FE-10 Mission & Operational Dashboard	Medium	16
Total Estimated Effort (man-days)			**184**

1.2 Project Objectives

Deliver a traceable UAV mission-operations platform for planning, pre-flight validation, real-time monitoring, incident management, post-flight review and AI-assisted risk assessment. AI uses a configured external OpenAI or Gemini API; the team implements context preparation, API integration, result display and error handling. Operators and Operations Managers review the advice; approval and operational decisions remain with authorized users. No model training or low-level autopilot algorithm development is planned. Flight preparation is implemented; aircraft command execution is future scope as specified below.

Quality objectives:

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Peer/Document Review	100% of baseline documents reviewed	≤ 8 major defects	≤ 20%	Track review findings and close before baseline
2	Unit Test	≥ 80% coverage for core backend/service logic	≤ 15 open defects	≤ 30%	Critical modules require unit tests

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
3	Integration Test	100% critical service integrations exercised	≤ 10 open defects	≤ 25%	MAVLink, MQTT/WebSocket, DB, AI and map integration
4	System Test	100% of 4 main flows + critical exceptions	0 Critical; ≤ 5 Major open	≤ 20%	Complete system tests by end W13; no Critical defect at council-ready release
5	Acceptance/Demo Test	100% planned end-to-end main flows demoable	0 Critical/Major blocking	≤ 5%	Council 1.1 readiness W14 and official defense W15; full package submitted end W13

• Milestone timeliness: ≥ 90% of planned milestones completed within the target week.• Critical-flow readiness: 100% of the four main flows demonstrable by system-test/council readiness.• Release quality: 0 open Critical defects and no Major defect that blocks an end-to-end main flow.• Traceability: all baseline FRs mapped to design, implementation, test evidence and demo where applicable.

The initial 184 feature man-days are retained as a provisional planning envelope: requirements/analysis 15%, architecture/design 15%, implementation/integration 40%, testing/quality 20%, and management/documentation/deployment 10%. FE-09 covers API integration, mission context, UI, failure handling and functional validation. These are effort estimates, not elapsed task durations or full-time allocations; the four-member team reviews capacity weekly and refines estimates after the SRS baseline.

1.3 Project Risks

Risks are reviewed weekly. High-impact risks are escalated to the team leader and supervisors, with mitigation actions assigned an owner and due date.

#	Risk Description	Impact	Possibility	Response Plans
1	MAVLink/simulator integration instability or inconsistent telemetry	High	Medium	Prototype early; use simulator test scenarios; isolate integration gateway; log raw telemetry.
2	Geospatial validation produces false positives/negatives	High	Medium	Define spatial rules and test datasets; unit-test waypoint/segment/polygon/altitude cases.
3	External AI API unavailable, rate-limited or over budget	High	Medium	KhanhGNG owns mitigation: prototype early; configure request limits/timeouts, monitor usage, protect server-side API keys and show clear failure/retry states. Core mission functions remain available.

#	Risk Description	Impact	Possibility	Response Plans
4	AI response is unsupported, incomplete or inconsistent with mission data	High	Medium	KhanhGNG owns mitigation: send relevant available data, identify missing context, validate response structure, show source mission data and require user review. Test prepared scenarios; AI does not approve missions.
5	Scope overload across Web, Mobile, telemetry, AI and analytics	High	Medium	Prioritize four main flows; freeze registration scope at end W6; Review 2 confirms required features within that scope; defer non-critical enhancements.
6	Team member unavailable / workload imbalance	Medium	Medium	Cross-review modules; maintain shared documentation; weekly rebalance via task tracker.
7	Integration defects discovered late	High	Medium	Prototype early; incremental testing from W7 Package 1; CI checks and regression through Packages 2 and 3; complete system-test dossier by end W13.
8	Deployment/network service failure before review/council	High	Low	Maintain reproducible deployment, local fallback/demo environment and pre-event readiness checklist.
9	W7 Package 1 deadline exceeds available student capacity	High	Medium	NghiaHQ coordinates MF1; reuse W2-W3 prototypes, complete preparatory design in W5-W6 and share FE-01 work. Review member availability weekly; escalate delivery risk early without moving mandatory submission deadlines.

2. Management Approach

The team uses incremental development around four main flows, weekly supervisor meetings, tracking and documented reviews. The controlling timeline is SWP490_Lo-trinh-Capstone_v1.0 (09/2026): Review 1 in W4-W5, Review 2 in W8-W9, Council 1.1 eligibility in W14 and Council 1.2 defense in W15. The SE Capstone Project Review Guideline supplies review content and minutes procedures; its Final Product review content is prepared for the council stage rather than scheduled as an additional W13 review.

2.1 Project Process

Four main flows are delivered through three iterations. Iteration 1 completes MF1 Mission Planning & Pre-flight Validation plus shared FE-01 by end W7 as Software Package 1. Iteration 2 in W10-W11 implements MF2 Real-Time Monitoring & Incident Management for Software Package 2. Iteration 3 in W12-W13 delivers MF3 Post-Flight Analysis & Mission History and MF4 AI-Assisted Mission Risk Assessment in parallel, then integrates and system-tests all four flows. Feasibility prototypes in W2-W3 and preparatory design in W5-W6 support the W7 code deadline; W8-W9 accommodates Review 2 and mandatory-feature feedback. Iteration count does not change the four-flow scope.

2.2 Quality Management

Quality is managed through prevention, peer review, automated/unit testing, integration testing, system testing and traceability. Definition of Done requires updated documentation, reviewed code, passing tests and demo evidence for each completed main flow.

- Defect Prevention: coding conventions, API contracts, architecture/design reviews, reusable validation rules, CI checks and early technical prototypes.
- Reviewing: peer review for requirements/design/code; supervisor review at weekly checkpoints; review feedback is logged and closed with evidence.
- Unit Testing: core backend services, geospatial validation, incident rules, AI context preparation, response validation and error handling using mocked provider responses.
- Integration Testing: simulator/UAV to gateway, telemetry persistence and real-time clients; validation to approval; incident/history/report integration; configured OpenAI or Gemini API success, timeout, rate-limit and malformed-response scenarios. Live API smoke tests complement repeatable mocked tests.
- System Testing: end-to-end execution of the four main flows plus critical exception scenarios, latency/reliability checks and regression tests.

2.3 Training Plan

Training is performed as targeted self-study and technical prototyping before the corresponding implementation increment. Knowledge is shared through short internal walkthroughs and code reviews.

Training Area	Participants	When, Duration	Waiver Criteria
MAVLink + UAV simulator integration	ThongLQ, NghiaHQ, KhanhGNG, DucDDA	W2–W3, 2–3 days	Waived if member can ingest and parse simulator telemetry end-to-end
Geospatial algorithms / geofence validation	NghiaHQ, KhanhGNG, DucDDA	W2–W5, 2 days	Waived after passing route/polygon/altitude prototype tests
External AI API integration and response handling	All members; KhanhGNG leads	W2-W3 prototype; W6 design, 2-3 training days	Waived after demonstrating mission-context request, valid response display, timeout/rate-limit handling and secure server-side API configuration
ASP.NET Core / Next.js / EF Core project conventions	All members	W2-W6, ongoing self-study	Waived after code review confirms team conventions
Flutter/Riverpod & real-time client updates	ThongLQ, NghiaHQ, KhanhGNG, DucDDA	W5-W7, 2 days	Waived after mobile/dashboard prototype
Testing, Git workflow & CI/CD	All members	W2-W6, 1-2 days	Mandatory baseline; practical check through PR + test pipeline

3. Project Deliverables

The plan follows the supplied September 2026 roadmap (pp. 2-5), using semester weeks as authoritative. Calendar dates assume W1 begins 07 September 2026; this assumption and all individual appointments require confirmation from university announcements. Current placeholders are Review 1 on 02 October (W4), Review 2 on 30 October (W8), Council 1.1 on 11

December (W14), and Council 1.2 on 18 December (W15). Common dossier deadlines are end W3, W7 and W13; defense documents are due at least two working days before the actual session. Where the guideline states 24 hours, this plan applies the stricter roadmap lead time and earlier common deadline. Adjust submission dates earlier if required by the actual appointment.

#	Deliverable	Due Date	Notes
1	Report 1 - Project Introduction	End W1	Scope, ten features, actors, context and four main flows
2	Report 2 + Project Schedule	End W2	Management plan, WBS, risks, responsibilities and schedule
3	Review 1 dossier	End W3	Registration; Reports 1-2; Report 3 overview (Product Overview, User Requirements, System Functional Overview); Project Tracking and review slides
4	Review 1 minutes / feedback	W4-W5	Review appointment TBA; agree/circulate minutes before session end; 48-calendar-hour feedback window
5	Confirmed registration / scope baseline	End W6	Update if required by Review 1; supervisor confirmation; scope frozen after W6
6	Review 2 dossier + Software Package 1	End W7	Full Report 3 SRS; Report 4 SDD; Report 5 Test Plan; Iteration 1 source/database scripts; updated Reports 1-2, Tracking and slides
7	Review 2 minutes / mandatory feature list	W8-W9	Requirements and design feedback; required features remain within the W6 registration scope
8	Software Package 2 + Reports 3-5 updates	End W11	Complete MF2 increment, source/database scripts, updated design and test records
9	Council 1.1 dossier / Full Software Package	End W13	Reports 1-6; complete Report 5 test cases/results/defect list; draft Report 6; registration, review minutes, Tracking and demo
10	Council 1.1 eligibility decision	W14	Three-member council checks product against registration, SRS, design and required feature list
11	Council 1.2 dossier / Report 7 / final package	At least 2 working days before defense	Report 7, defense slides, source/database scripts, testing documents and Tracking. Provisional submission Tue 15 Dec for Fri 18 Dec defense
12	Council 1.2 - official defense	W15	Conditional on passing Council 1.1; all members present/demo/respond
13	Weekly Report / Project Tracking	Every week	GVHD meeting; WBS, Issues, Defects, Q&A, risk/capacity and feedback tracking
14	Council 2 recovery package (conditional)	About W19; date TBA	Only after failure at Council 1.1 or 1.2; next semester, about four weeks after W15; revised product/docs and feedback-response matrix; submit at least 2 working days before session

3.1 High-Level 15-Week Schedule

Week	Phase / Gate	Key Activities	Primary Output
W1	Initiation	Registration, supervisor coordination, scope and context	Report 1 by end W1
W2	Planning / Feasibility	PMP, Schedule, risks and technical prototypes	Report 2 + Schedule by end W2
W3	Requirements Overview	Report 3 overview, Tracking and Review 1 slides/dossier	Review 1 dossier submitted by end W3
W4	Review 1 window	Provisional review Fri W4; official window W4-W5; record minutes	Feasibility/scope feedback
W5	Scope refinement / preparation	Review 1 if assigned W5; feedback; SRS, architecture and MF1 design	Updated requirements/design drafts
W6	Scope freeze	Resolve scope feedback; supervisor-confirmed registration; design preparation	Registration frozen by end W6
W7	Iteration 1 / Review 2 submission	Complete MF1/shared FE-01; finalize SRS, SDD and Test Plan	Package 1 + Review 2 dossier by end W7
W8	Review 2 window	Provisional review Fri W8; official window W8-W9	Minutes, design feedback and required features
W9	Review 2 / feedback closure	Review if assigned W9; baseline feature list and revise backlog	Updated SRS/SDD/Test Plan within frozen scope
W10	Iteration 2	MF2 gateway, persistence and monitoring	Integrated monitoring build
W11	Iteration 2 completion	MF2 incidents/alerts; tests and Reports 3-5 updates	Software Package 2 by end W11
W12	Iteration 3	MF3 post-flight and MF4 AI API in parallel; integrate; draft user guides	Internal feature-complete build
W13	System tests / Council submission	System tests, defects, complete Report 5 and draft Report 6; full package	Council 1.1 dossier by end W13
W14	Council 1.1 - Eligibility	Three-member council; product/document review; prepare defense draft	Eligibility decision and feedback
W15	Council 1.2 - Defense	Close feedback; submit dossier at least 2 working days before defense; rehearse	Report 7, final product and official defense if eligible

4. Responsibility Assignments

Responsibility codes: D = Do/Owner; R = Review; S = Support; I = Informed. Team-confirmed allocation (16 September 2026): MF1 NghiaHQ; MF2 ThongLQ; MF3 DucDDA; MF4 KhanhGNG. Each owner coordinates requirements, design, implementation and testing for the assigned flow; other members provide support as needed. Existing cross-cutting responsibilities follow the agreed allocation: KhanhGNG coordinates Web frontend, GitHub, database/deployment work; NghiaHQ coordinates backend integration and project tracking. Resource Names identify contributors rather than full-time bookings.

D = Do/Owner; R = Review; S = Support; I = Informed; blank = Omitted.

Responsibility	ThongLQ	NghiaHQ	KhanhGNG	DucDDA	Supervisors
Project Planning & Tracking	S	D	S	S	R
Report 1 / PMP / Final Report coordination	D	S	S	S	R
Requirements & SRS	S	D	S	S	R
Architecture & Backend Integration	S	D	S	S	R
MF1 - Mission Planning & Pre-flight Validation	S	D	S	S	I
MF2 - Real-Time Monitoring & Incident Management	D	S	S	S	I
MF3 - Post-Flight Analysis & Mission History	S	S	S	D	I
MF4 - AI-Assisted Mission Risk Assessment	S	S	D	S	I
Web/Map UI	D	S	S	S	I
Mobile / Dashboard / DevOps	D	S	D	S	I
Test Planning & System Testing	S	D	S	D	I
Deployment / Demo / User Guide	D	S	S	S	R

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Weekly GVHD meeting / Weekly Report	All members + supervisors	Review progress/capacity; update WBS, Issues, Defects and Q&A	Weekly W1-W15	Meeting, Project Tracking and Project Weekly Report
Daily team coordination	All members	Task progress, blockers and shared implementation	Working days	Team chat + task board
Pull Request / Code Review	Developers + reviewer	Source quality and reproducible evidence	Per PR	Repository PR, review comments and CI
Review/Council dossier submission	Team to supervisors/reviewers/council	Send registration, slides and required documents	End W3/W7/W13; defense >=2 working days before	Use the announced channel; apply earlier deadline if actual appointment requires it
Review meeting / minutes	All members, GVHD and reviewers	Present; agree checklist/findings; record and circulate minutes	W4-W5 and W8-W9; minutes before session end	Prescribed minutes and email procedure; retain evidence

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Supplementary review feedback	Team, GVHD and reviewers	Resolve or record additions to agreed minutes	Within 48 calendar hours after review	Reply to review minutes thread; archive correspondence
Council sessions / feedback	All members + council	W14 eligibility; W15 defense if eligible	W14 / W15, appointments TBA	Demo, presentation, minutes and feedback-response tracking
Risk/Issue Escalation	Owner -> leader -> supervisors	Resolve high-impact delivery issues	As needed; within 24h for high impact	Issue tracker + direct communication

6. Configuration Management

6.1 Document Management

Maintain English Reports 1-7 using course templates, Project Schedule, Project Tracking and Weekly Reports in the shared document workspace with version labels and archived baselines. Freeze the supervisor-confirmed registration at end W6; Review 2 records mandatory features within that scope. Retain review/council minutes, feedback correspondence and evidence; update the Record of Changes and traceability after each gate. Submit common dossiers at end W3, W7 and W13 and the defense dossier at least two working days before the actual appointment. Only groups failing Council 1.1 or 1.2 prepare the conditional Council 2 package with a feedback-response matrix about four weeks after W15.

6.2 Source Code Management

Source code is managed in Git. Work is performed on feature/fix branches and merged through pull requests after review. The main/develop branch policy, commit naming, issue references and release tags are agreed by the team. Secrets are excluded from source control and provided through environment configuration. Review/council builds are tagged so they can be reproduced.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	Web frontend: Next.js; backend: C# / ASP.NET Core; existing mobile scope: Flutter/Dart (Riverpod); MAVLink; MQTT/WebSocket; configured OpenAI or Gemini API called from the backend.
Database / Data Access	Supabase PostgreSQL; Entity Framework Core for backend data access; mission, telemetry, incident and report persistence. ThongLQ coordinates database/deployment work; NghiaHQ coordinates backend integration.
IDEs/Editors	Visual Studio / Visual Studio Code; Android Studio as needed
Diagramming	Draw.io / diagrams.net, PlantUML or equivalent
Documentation	Microsoft Word/Excel, Google Drive/Docs/Sheets
Version Control	GitHub repository with feature branches, pull requests and tagged baselines; ThongLQ coordinates repository work.
Deployment / CI	Deployment and CI build/test pipeline coordinated by ThongLQ; environment configuration keeps credentials outside source control.

Category	Tools / Infrastructure
Testing / Simulation	MAVLink-compatible UAV simulator/autopilot environment; unit/integration/system test tools

Review 2 management update — 19 October 2026

This update records preparation work, not a declaration that all functions are delivered. Review 2 dates in the original schedule remain provisional; the actual university appointment must be confirmed. The guideline requires the dossier at least 24 hours before review; use an earlier formally confirmed course deadline if applicable.

Workstream	Owner from existing PMP	Review 2 action / acceptance
MF1	Ha Quang Nghia	Implement/review pending-update staging and reject-preserves-approved behavior; verify scope and persisted validation.
MF2	Le Quoc Thong	Verify telemetry source; complete incident action/detector and FE monitoring integration.
MF3	Dang Dao Anh Duc	Verify report aggregation, coverage, authorization and execution linkage.
MF4	Giang Nguyen Gia Khanh	Own AI context, provider integration and failure handling. ThongLQ retains dossier/CI coordination.
All members	Peer reviewers	Review own/assigned vs managed scope, test results and actual demo evidence; confirm final SRS decisions.

Risk	Mitigation / evidence
Pending update overwrites approved mission	Staging + atomic approval + regression cases M03/M04.
Registered AI scope differs from implementation	Supervisor decision recorded separately; retain registration unchanged.
MAVLink gateway / realtime demo unavailable	Prepare deterministic recorded fixture, labelled playback; never present it as real flight.
SRS finalization items unresolved	Decision checklist with owner/response before submission; reviewer decisions are recorded in the designated confirmation fields.
No live test evidence in this dossier	Five integration cases have recorded Pass results with build identity and raw logs in Report 5/test-evidence; staging acceptance is tracked separately.

Review 2 completion gates

Peer-review SRS → confirm scope/open decisions → review SDD diagrams and DB migrations → run tagged demo/test build → capture results → submit registration/PMP/SRS/SDD/Test Plan/slides before deadline → record and agree minutes during review. Send minutes as required by the guideline; communication actions follow the team's submission procedure.

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

Delivery and accountability

ThongLQ owns the MF2 gateway/readiness increment and its integration evidence; NghiaHQ reviews planning/hash/preflight dependencies; KhanhGNG owns MF4 AI risks and provider tests. Current schedule dates and effort totals are historical planning estimates, not evidence that command execution is implemented. Plan SITL/upload/dispatch work separately before committing execution dates.